

Part K: Reflection on the Design Project

Defining the Problem

What Went Well:

Mrs. Cantrell approached us directly with a clear and well-defined problem. This early clarity allowed us to jump straight into brainstorming with a full understanding of the issue and its requirements.

What Didn't Go Well:

There was minimal opportunity for errors during this stage since the problem was pre-defined. This part of the process went smoothly and efficiently.

What Would We Do Differently:

Given the clarity provided from the start, there is nothing we would change. The efficiency of this phase set a solid foundation for the rest of the project.

Brainstorming

What Went Well:

Our group's pre-existing familiarity with each other fostered open communication and respect for each other's ideas. This allowed us to freely share and build upon our concepts, ultimately enriching the design process.

What Didn't Go Well:

We spent an excessive amount of time during this phase, attempting to fully explore and analyze each idea. While thorough, it led to delays in other stages.

What Would We Do Differently:

In future projects, we would implement a stricter timeline during brainstorming to avoid bottlenecks and improve time management. Early realization of this would have benefited the later stages.

Idea Selection

What Went Well:

Mrs. Cantrell was actively involved in this process, providing direct feedback that guided our design choices. Her input helped streamline the decision-making process.

What Didn't Go Well:

Like the problem definition, this stage was straightforward with minimal complications.

What Would We Do Differently:

With helpful guidance from Keith Stiltner, an avid outdoorsman and engineer, there is nothing we would change. His feedback was instrumental in shaping the final design.

Prototype Design

What Went Well:

With a clear vision in mind, designing the prototype was efficient. The understanding that the shoe needed to be raised drove the design choices effectively.

What Didn't Go Well:

There were no major setbacks in this phase. The design process proceeded as planned.

What Would We Do Differently:

Nothing needed to change. The planning and execution during this phase were both smooth and effective.

Prototype Building

What Went Well:

It took three iterations to reach the final design, which we were satisfied with. Testing began on the third prototype, highlighting our progress.

What Didn't Go Well:

We underestimated the time required for building. What was planned to take one week stretched to three, partially due to factors beyond our control.

What Would We Do Differently:

In future projects, we would enforce a more rigid construction schedule to prevent delays and optimize project flow.

Testing

What Went Well:

The testing phase provided valuable insights into both the strengths and weaknesses of our design. We also successfully identified the optimal teaching method to accompany the shoe.

What Didn't Go Well:

The 3D printing process was time-consuming. Each stand took approximately 76 hours to complete, limiting our output.

What Would We Do Differently:

If resources permitted, larger-scale testing would be ideal, especially for validating the instructional method. More 3D printers or a larger budget would have greatly improved our efficiency.

